

Signals from Abroad: Climate Policy Appropriateness and Informational Spillovers*

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Abstract

Addressing climate change requires politicians and citizens to assess whether policy action is appropriate given underlying risks. This paper develops a formal theory of domestic and international climate policymaking centered on how politicians, anti-climate interest groups, and voters interact under uncertainty about the severity of climate change. I show that foreign climate actions influence domestic political support for reform because voters use them as information about whether climate policy is warranted. These informational spillovers create international political interdependence even when climate politics is rooted in domestic distributive conflict and absent the collective action concerns emphasized by existing theories of global climate cooperation. However, the domestic political consequences of foreign climate action depend on the information available to citizens. Anti-climate interests strategically shape this informational landscape through public messaging, and their optimal communication strategy varies with public expectations of climate severity: special interests engage in uninformative messaging when concern is low but counterintuitively acknowledge climate risks once concern is high. Foreign climate actions are thus most informative of the need for climate action when special interest messages are uninformative, precisely when domestic informational conditions are least favorable to reform. Descriptive evidence on fossil fuel industry messaging and European public opinion is consistent with the theory's core informational mechanism.

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Formulating policies to combat climate change requires politicians and citizens to assess whether climate risks warrant policy action (Balcazar and Kennard 2025; Gazmararian and Milner 2025; 2026). In forming these assessments, publics rely on informational cues about the need for climate reforms, including both the messages disseminated by special interest groups (Kollman 1998; Stokes 2020; Schnakenberg and Turner 2024) and the climate policies adopted by other countries (Tingley and Tomz 2014; Gaikwad, Genovese and Tingley 2025). How these signals shape domestic support for climate action is therefore central to understanding both domestic climate policymaking and international climate cooperation.

This paper develops a formal theory of domestic and international climate policymaking centered on the trilateral relationship among politicians, special interests, and voters under uncertainty about climate risks. Politicians, with an eye toward reelection, determine climate policy anticipating that voters will judge their choices by inferring whether reforms were warranted given the expected severity of climate change. Special interests, seeking to deter climate action, invest in public messaging that shapes what voters can learn about climate risks. Voters assess the *appropriateness* of domestic climate policy by using information from both special interest messaging and foreign climate actions.

The model produces several novel insights about how information shapes domestic and international climate politics. The central result is that foreign climate actions influence domestic political support for reform because voters treat them as evidence about whether climate policy is warranted at home. In a “second image reversed” fashion (Gourevitch 1978), climate policies adopted abroad generate *informational spillovers* that link domestic climate politics across countries. However, the political consequences of foreign climate action are conditional. Foreign climate policies provide valuable information when domestic signals about climate risks are weak or distorted, but become substantially less informative once citizens can assess climate severity more directly.

These international dynamics emerge from a domestic political process involving politi-

cians, voters, and special interests. I demonstrate that the messaging strategies of anti-climate special interests evolve sharply with public expectations about climate severity. Although these groups have long had access to accurate climate models and forecasting ([Oreskes and Conway 2011](#); [Supran and Oreskes 2021](#); [Supran, Rahmstorf and Oreskes 2023](#)), they face a strategic choice about whether to honestly report these findings or to “misreport,” downplaying the climate threat to discourage support for reforms. When voters perceive climate risks to be limited, special interests engage in maximal misreporting, issuing communications that understate expected environmental harms. As public perceptions of climate severity rise and climate action becomes broadly viewed as warranted, however, special interests instead shift toward unbiased messaging. Counterintuitively, special interests then shift to unbiased messaging: by acknowledging severe climate risks, the interest group can constrain politicians via the threat of electoral removal following incommensurate policymaking.

The informational mechanism developed here reconciles international climate politics with distributive accounts of climate policymaking (e.g., [Aklin and Mildenberger 2020](#); [Colgan, Green and Hale 2021](#); [Ross 2025](#); [Gazmararian and Tingley 2026](#)). The model features distributive conflict: special interests consistently oppose climate reforms while publics support them if viewed as warranted. Nevertheless, international climate actions influence domestic politics because voters use foreign policy choices as informational cues when evaluating climate policy appropriateness at home. This mechanism departs from standard theories of international climate cooperation, which assume climate policies are strategically complementary through coordination incentives ([Barrett 2016](#); [Keohane and Victor 2016](#); [Hale 2020](#)) or substitutable through free-riding incentives ([McAllister and Schnakenberg 2022](#); [Kennard and Schnakenberg 2023](#)). Instead, whether foreign climate action raises or dampens domestic reform depends on how voters interpret the information available to them.

I evaluate the theory’s empirical implications with descriptive evidence on both special interest climate messaging and public opinion toward climate policy. First, I trace the evo-

lution of public communications of major fossil fuel interests such as ExxonMobil, shifting from strategies exploiting climate uncertainty to decrease support for reform toward greater acknowledgment of climate risks and support for limited action ([Dunlap and McCright 2011](#); [Williams et al. 2022](#)). This trajectory is consistent with the theory’s expectation that misreporting becomes less pervasive as public perceptions of climate severity rise. Second, using Eurobarometer surveys and data on climate law adoption, I examine whether beliefs about climate seriousness vary systematically with domestic climate action, foreign climate action, and special interest messaging in the manner implied by the theory. The most distinctive pattern concerns the informational value of foreign climate action. Foreign climate policies influence beliefs about climate seriousness when signals about climate severity are noisy, but attenuate in informativeness once such signals become available. These findings are consistent with the theory’s claim that international informational spillovers are conditional on what voters can learn about climate severity at home, operating most strongly when citizens lack direct information about climate risks. Moreover, such patterns are difficult to reconcile with collective action accounts of global climate cooperation, as observed effects operate through shifts in public perceptions rather than through changes in countries’ direct strategic valuations of foreign climate policy.

This paper is closely related to a body of formal work on the political economy of climate policymaking and on electoral accountability in international politics. Formal accounts of climate cooperation have studied issues of strategic interdependence from the perspective of collective action problems ([Kennard and Schnakenberg 2023](#)), designing optimal agreements with heterogeneous domestic commitments ([McAllister and Schnakenberg 2022](#)), and domestic political competition ([Battaglini and Harstad 2020](#); [Melnick and Smith 2025](#)). The present paper develops a complementary account in which cross-country interdependence emerges from how domestic publics interpret the value of foreign climate actions, with coordination and free-riding incentives emerging endogenously. The paper also builds on formal

models of electoral accountability in international politics, in which informational asymmetries between voters and politicians shape policy outcomes in domains ranging from the politics of war (Schultz 2005; Tarar 2006) to foreign policy appointments (Malis 2024). While these accounts treat what voters can learn as exogenously given, the theory here endogenizes it: the information voters have depends on strategic information dissemination by an interest group, which shapes how international politics affects domestic policymaking.

The paper’s broader contribution lies in demonstrating that international political dynamics may shape domestic climate policymaking through informational channels that extend beyond traditional cooperative mechanisms. A growing literature argues that climate politics is primarily structured by domestic distributive conflict (Aklin and Mildenberger 2020; Ross 2025) rather than by the collective action problems emphasized in canonical theories of international environmental cooperation (Barrett 2003; Stavins 2011; Kennard and Schnakenberg 2023). This perspective has substantially improved our understanding of the domestic political economy of decarbonization,¹ but leaves open the question of whether and how international politics still matters for domestic climate policymaking. The theory developed here takes up that question by identifying an informational mechanism through which foreign climate action directly influences domestic political incentives, even in the absence of formal coordination mechanisms or collective action concerns.

The model also contributes to literatures on the mass electoral and attitudinal implications of climate policy. Observational evidence on whether voters reward or punish climate policies at the ballot box is mixed (e.g., Stokes 2016; Urpelainen and Zhang 2022; Bolet, Green and González-Eguino 2024; Colantone et al. 2024; Gazmararian 2025; Voeten 2025). The theory developed here helps unpack this empirical ambiguity: whether climate action elicits electoral reward or punishment depends on what voters expect *ex ante* about climate

¹Extant studies point to, *inter alia*, electoral institutions and incentives (Finnegan 2022; Melnick 2024), special interest influence (Mildenberger 2020; Stokes 2020), and sectoral conflicts (Aklin and Urpelainen 2013; Cheon and Urpelainen 2013; Hughes and Urpelainen 2015) as drivers of climate policymaking.

severity, so the same policy can be electorally costly in one context and electorally beneficial in another. A related body of work shows that citizens’ support for climate policy responds to the actions of other countries (e.g., [Li et al. 2004](#); [Bechtel and Scheve 2013](#); [Tingley and Tomz 2014](#); [Bernauer and Gampfer 2015](#); [Bernauer et al. 2016](#); [Beiser-McGrath and Bernauer 2019](#); [Bayer and Genovese 2020](#); [Gaikwad, Genovese and Tingley 2025](#)), often interpreting foreign action as altering the strategic value of domestic cooperation. The theory here implies that this responsiveness should itself be conditional, operating through what voters can infer about climate severity and therefore depending on the signals they receive about climate risks.

I also shed light on the role of “outside lobbying” ([Kollman 1998](#)) in climate politics. Public messaging may promote doubt or denialism to delay climate action (e.g., [Oreskes and Conway 2011](#); [Frumhoff, Heede and Oreskes 2015](#); [Supran 2022](#)), complementing lobbying and contribution efforts ([Brulle 2014](#); [Kim, Urpelainen and Yang 2016](#); [Brulle 2018](#); [Stokes 2020](#); [Brulle 2021](#); [Cory, Lerner and Osgood 2021](#); [Schnakenberg and Turner 2024](#)). This paper isolates how special interests strategically design public messaging to shape beliefs about the necessity of climate action, thereby influencing climate policymaking through electoral incentives. Notably, the theory proposes a rationale for why interest groups may acknowledge climate risks, and why such messaging may nevertheless deter climate action.

More generally, this paper relates to a literature on the political consequences of cross-national observation of policy choices (e.g., [Simmons, Dobbin and Garrett 2006](#); [Gilardi 2012](#)). I identify a distinct route through which foreign policy choices shape domestic politics: voters use foreign behavior as evidence about whether domestic policy action is warranted. Even where countries do not face direct policy impacts from cooperating, foreign action enters domestic politics inferentially, with consequences for politicians’ electoral fortunes and the likelihood of policy adoption at home. These dynamics suggest the value of attending to how domestic publics interpret what happens abroad, and how the influence of foreign actions

depends on the cues voters receive about the severity of climate risks.

Model

The model studies how uncertainty about climate risks shapes domestic and global climate politics. Politicians must decide whether to pursue climate reforms when voters are uncertain whether such reforms are warranted. Voters rely on informational cues to assess the appropriateness of climate policy, including both messages disseminated by special interests and climate policies adopted abroad. Special interests strategically shape what voters can learn about climate severity in an effort to discourage reform.

Accordingly, the model studies two countries, i and j . Each nation contains a politician P (“she”) and a representative or median voter V (“he”).² There is also a special interest group S (“it”) that seeks to influence policy in both nations.

There are two policy-relevant states of the world $\omega \in \{0, 1\}$. In simplified terms, ω represents the severity of climate change’s effects or the vulnerabilities to climate-related damages. Each state of the world carries a “correct” or “appropriate” policy response that is commensurate with anticipated environmental harms: state $\omega = 1$ indicates a scenario in which greater climate policy reforms are appropriate because climate change’s effects are more severe, while the case of $\omega = 0$ represents an instance in which the status quo is sufficient. As will be detailed below, players have policy preferences that depend on this underlying state. The true value of ω is unobserved, but players share a common prior $P(\omega = 1) = \pi \in (0, 1)$, capturing the expected severity of climate change.³

The game begins with the special interest group committing to the design of information

²The language of voters and elections is used to ease exposition, but the model need not be scoped to democratic countries. In nondemocratic states, this player may be an elite or other individual whose political support is pivotal for the leader’s survival in office.

³While individual knowledge about climate change may vary globally (Lee et al. 2015; Kennard 2025), the prior represents a common baseline from which all actors have expectations about climate change’s effects, which may stem from common informational sources like the IPCC (Hai 2025).

about the state of the world. This takes the form of designing a signal s with distribution $P(s|\omega)$. This signal represents a report about climate change’s severity, which takes on two values, $s \in \{0, 1\}$. Since I focus on the case of an anti-climate interest group like ExxonMobil,⁴ the group seeks to convince domestic publics that $\omega = 0$, implying that maintaining the status quo is the correct policy response. Given the preferences of the interest group, as well as the dichotomous nature of the state of the world, the choice of a signal distribution can be written as

$$P(s = 0|\omega = 0) = 1. \quad P(s = 1|\omega = 0) = 0.$$

$$P(s = 0|\omega = 1) = \beta. \quad P(s = 1|\omega = 1) = 1 - \beta.$$

Whenever the true state is $\omega = 0$, the group will always send the signal $s = 0$: it would never be in the group’s interest to communicate that climate change poses a threat that demands action when the correct policy aligns with its preferences for inaction. However, if $\omega = 1$, there is some probability $\beta \in [0, 1]$ that the special interest reports signal $s = 0$. I will therefore refer to β as the level or intensity of “misreporting” about the true effects of climate change. Higher values of β mean that the special interest is more likely to send the message that climate change warrants minimal action when the true state would beget greater reforms. The signal structure implies that the choice of β is isomorphic to the choice of the distribution itself, so the special interest chooses β optimally in order to maximize the chances that politicians enact policy congruent with $\omega = 0$; the group receives a payoff of 1 for each country that does not act and zero otherwise.

After the special interest group chooses misreporting β , which is observed by all players,

⁴Concentrating on a single interest group allows for a concrete understanding of the incentive structure for misreporting, but one could also interpret s as the “net messaging” a voter receives from multiple groups. I assume the group is anti-climate, which captures the empirical regularity of anti-climate lobbying, especially in the United States (Dunlap and McCright 2011; Brulle 2014; Dunlap and McCright 2015). See Appendix A for an extension with a pro-climate group.

the politicians in each country take a policy action on climate change, $a_i \in \{0, 1\}$. The action $a_i = 1$ represents broad climate reform or more intensive policies that might regulate the production of fossil fuels, and $a_i = 0$ captures the status quo or minimal policy measures.⁵

While climate change’s expected effects remain uncertain, the politician has an informational advantage over the voter because she observes a signal about the state, indicating the necessity of potential climate reforms. The politician’s signal is $x_i \sim G(\cdot; \omega)$, where the distribution function $G(x; \omega)$ admits a log-concave density $g(x; \omega)$ with the monotone likelihood ratio property and is state-symmetric⁶ (e.g., the normal distribution, $x_i \sim N(\omega, \sigma)$). I assume signals are sufficiently precise such that a unique equilibrium is ensured (e.g., if $x_i \sim N(\omega, \sigma)$, assume that σ is sufficiently small; see Lemma A.8 for more discussion). Politicians’ signals x_i and x_j are conditionally independent given ω .

Politicians and voters share aligned policy preferences: both prefer that policy matches the state, $a_i = \omega$, or that climate policy is appropriate. When this occurs, both receive a policy payoff of 1. However, since voters cannot directly observe ω , they must instead infer whether incumbent politicians acted appropriately. Once policies are implemented, voters observe the triple (a_i, s, a_j) , consisting of the domestic policy choice, the special interest group’s message, and the foreign country’s action. Based on these observables, voter i forms a posterior belief about the incumbent leader’s appropriateness, $\mu_i(a_i, s_i, a_j) = P(\omega = a_i | a_i, s_i, a_j)$, and either retains or replaces the politician, $r_i \in \{0, 1\}$. The politician receives a payoff of 1 if retained.

If the voter retains the incumbent, he receives 1 if the policy is appropriate, i.e., if $a_i = \omega$, and 0 otherwise. If he replaces the incumbent, his payoff is a random draw $\varepsilon_i \sim F(\cdot)$ where $F(\cdot)$ is a distribution function with a log-concave density. This payoff could represent the

⁵Explicitly modeling costs to climate action would only bias the results toward $a_i = 0$ and does not qualitatively alter the equilibrium intuitions or mechanisms at play.

⁶State symmetry assumes that there exists a “center” c such that $g(c+x; 1) = g(c-x; 0)$ for any x . Most results only require MLRP but state symmetry greatly facilitates exposition, simplifies results, and allows for comparisons across domestic and international politics.

probability that a challenger pursues appropriate climate policy in the future, or the value of the incumbent politician on all other electorally salient dimensions that are independent of climate policy. The shape and support of $F(\cdot)$ modulate how much the voter cares about climate policy relative to other issues, capturing salience as well as structural electoral factors such as partisan asymmetry and biases or incumbency advantages.⁷

For the special interest group and players in country i (country j 's are analogous), payoffs are formalized as follows:

$$\begin{aligned} u_S &= 2 - a_i - a_j. \\ u_P &= a_i\omega + (1 - a_i)(1 - \omega) + r_i. \\ u_V &= r_i(a_i\omega + (1 - a_i)(1 - \omega)) + (1 - r_i)\varepsilon_i. \end{aligned}$$

The timing of the game is summarized as follows:

0. Nature randomly draws the state ω .
1. The interest group chooses misreporting level $\beta \in [0, 1]$.
2. Politicians observe signals (x_i, x_j) and choose climate policies, $(a_i, a_j) \in \{0, 1\}^2$.
3. The interest group's message $s \in \{0, 1\}$ is realized. Voters form posterior beliefs $(\mu_i(a_i, s, a_j), \mu_j(a_j, s, a_i)) \in [0, 1]^2$ and retain or replace politicians, $(r_i, r_j) \in \{0, 1\}^2$.

I examine Perfect Bayesian equilibria. A strategy for the special interest group is a misreporting level $\beta \in [0, 1]$. A strategy for politician i is a function $\sigma_i : \mathbb{R} \rightarrow [0, 1]$ mapping her private signal x_i to a probability of choosing $a_i = 1$. A strategy for voter i maps the

⁷The empirical literature documents that individuals might display partisan motivated reasoning when learning about climate change ([Druckman and McGrath 2019](#)). One could imagine that there are different distributions of $F(\cdot)$ with different supports depending on whether the voter is ideologically aligned with the incumbent politician or not, but I leave this unmodeled, since, given a prior, motivated reasoning may be observationally equivalent to Bayesian updating ([Little 2025](#)). The model only requires that the politician's reelection probability to be increasing in the voter's belief that she took the correct action.

observed history (a_i, s, a_j) into a retention rule, $r_i : \{0, 1\} \times \{0, 1\} \times \{0, 1\} \rightarrow [0, 1]$. Voter i 's posterior beliefs are formed by Bayes's Rule on the equilibrium path.

Comments on the Model

The model makes several simplifying assumptions that warrant further interpretation.

The special interest's information structure. The special interest's information structure relies on symmetric uncertainty and an *ex ante* choice of a signal distribution, building on the framework of [Kamenica and Gentzkow \(2011\)](#).⁸ In this context, the commitment assumption allows for the formal analysis of the impact of the special interest's message on politician behavior, which is necessary if we think that a special interest has strict preferences over policy outcomes. This modeling approach also captures the indirect nature of outside lobbying: the special interest group seeks to prevent politicians from implementing climate reforms, and does so indirectly by targeting what the public can learn about climate severity. By manipulating the signal s , the group shifts the voter's posterior beliefs about the state of the world, subsequently shifting voters' electoral assessments of politicians.

However, rather than interpreting this as a literal "commitment to a disclosure rule," a usually restrictive assumption ([Little 2023](#)), one can interpret the group's choice as an *ex ante* investment in public relations and evidence suppression. Substantively, if the true state requires climate action, $\omega = 1$, internal scientists and researchers may generate evidence unfavorable to the group's preferences. The parameter β represents the special interest group's structural capacity to suppress this whistleblowing, manufacture doubt, and successfully communicate a "safe" narrative to the public. If the group successfully suppresses the evidence, with probability β , the public receives the favorable signal $s = 0$. If the suppres-

⁸Unlike standard Bayesian persuasion models where the receiver of the signal directly takes the action, this setup separates the belief-updater (voter) from the decisionmaker (politician). The sender's (interest group) message operates indirectly, persuading the voter to electorally discount climate action, which subsequently filters upward to constrain the politician's policy choice.

sion apparatus fails, with probability $1 - \beta$, the truth about climate damages leaks, and the public observes $s = 1$. Symmetric uncertainty ensures there is no signaling embedded in the ex-ante choice of β .

Strategic adjustments. Models of international climate cooperation often assume that nations’ actions are substitutable, creating a free-rider problem (e.g., [Kennard and Schnakenberg 2023](#)), or that they are complementary because there is some international institution seeking to coordinate policy (e.g., [Barrett 2016](#)). The model does not explicitly assume either strategic complementarity or substitutability in countries’ climate policies. Rather, as will be shown, these strategic effects emerge endogenously in the model as a result of how voters perceive the appropriateness of foreign climate reforms and whether incumbent leaders are rewarded for matching or diverging from foreign behavior. Hence, in this model, international action shapes climate policy precisely through its effects on domestic politics, rather than through assumed strategic structure at the international level.

A natural concern is that the model omits any direct effects of policy externalities on countries’ propensities to take climate action. The aim of the theory is to isolate the informational channel through which foreign action shapes domestic politics, and adding a standard externality term would obscure rather than illuminate this mechanism (cf. [Paine and Tyson 2020](#)). The model should therefore be understood as characterizing one mechanism, voter inference about policy appropriateness, that operates alongside extant arguments that emphasize collective action and coordination problems.

Domestic politics. Similar to extant electoral accountability models (e.g., [Canes-Wrone, Herron and Shotts 2001](#); [Maskin and Tirole 2004](#); [Schultz 2005](#); [Malis 2024](#)), the animating force behind the domestic political interaction is an informational asymmetry between the politician and the voter. This generates an adverse selection problem: the voter cannot directly assess whether the politician’s chosen policy is appropriate. The present model

differs from this tradition, however, in the form of the asymmetry. In canonical accountability models, voters are uncertain about politician type—competence, congruence, or ideological alignment—and the voter’s task is to screen politicians who have favorable characteristics from those who do not. Here, by contrast, politicians are *ex ante* identical, and the voter seeks to simply reelect politicians who are more likely to have chosen appropriate policy.

Interpretation of the state, actions, and the politician’s signal. There is a connection between appropriate policy responses and the state of the world, which generates distributional conflict between the special interest, which has state-independent preferences, and the politician and voter, who have state-dependent preferences over policy. By way of interpretation, policy $a_i = 0$, the preferred choice of the special interest regardless of the state, might typify minimal climate reforms or consumer-facing policies that still allow for the combustion of fossil fuels, including policies that invoke the “individualization of responsibility.” Policy $a_i = 1$ encompasses more comprehensive reform or actions more likely to affect the production of fossil fuels. The politician’s signal x_i captures variation in the ability to identify the appropriate policy response. This may arise from differences in bureaucratic capacity, the quality of available scientific knowledge, or sensitivity to voters’ willingness to pay for climate policy given their beliefs about environmental risks.

Analysis

In what follows, I build the model sequentially to examine how each of the three sources of information—the domestic politician’s action, the special interest’s message, and the foreign politician’s action—affects climate policymaking. As will be shown, each of these actions is associated with a key mechanism—pandering, persuasion, and international cooperation. Let $\mathcal{I}$ be the set of information that the voter uses to form his assessment of appropriateness.

Additionally, it is useful to state up front that the equilibrium takes the following form:

1. The special interest chooses optimal misreporting $\beta^* \in [0, 1]$.
2. A unique, (symmetric) cutoff $\tilde{x}^*$ exists such that a politician in country i chooses policy $a_i = 1$ in state ω given signal x_i with probability $\sigma^*(x_i) = 1 - G(\tilde{x}^*; \omega) \in [0, 1]$.
3. Given $\mathcal{I}$, the voter in country i forms posterior belief about politician appropriateness $\mu_i(\mathcal{I}; \tilde{x}^*)$ and retains the politician with probability $F(\mu_i(\mathcal{I}; \tilde{x}^*))$.

Effects of Domestic Climate Action

First suppose that the only piece of information that the voter can observe is the politician's choice to engage in climate reforms at home. This baseline setting isolates how electoral incentives generate *pandering*. Pandering refers to the politician's tendency to choose the policy that the voter is *ex ante* more likely to perceive as appropriate in order to improve reelection prospects (Canes-Wrone, Herron and Shotts 2001; Maskin and Tirole 2004). Proposition 1 summarizes the core insights of the mechanism; proofs for all formal results can be found in Appendix A.

Proposition 1 *Let $\mathcal{I} = \{a_i\}$. There exists a unique equilibrium in which:*

- *the probability of climate action is increasing in π ;*
- *there exists a $\pi^\dagger$ such that the voter is more likely to reward $a = 0$ when $\pi < \pi^\dagger$ and more likely to reward $a = 1$ when $\pi > \pi^\dagger$;*
- *the politician panders to $a = 0$ when $\pi < \pi^\dagger$ and panders to $a = 1$ when $\pi > \pi^\dagger$.*

In this stylized setting, the voter evaluates the politician using only observed domestic policy choices and reelects the incumbent when climate policy appears justified given

expected climate risks. Let $\mu(a) = P(\omega = 1|a)$ be the voter's posterior belief about the politician's appropriateness given the policy choice, such that the probability the politician is reelected is simply $F(\mu(a))$.

Although the politician is better informed than the voter, she does not perfectly observe the true severity of climate risks. She therefore forms beliefs about whether climate reform is warranted given her private signal x . This posterior belief is $\eta(x) = P(\omega = 1|x) = \frac{\pi g(x;1)}{\pi g(x;1) + (1-\pi)g(x;0)}$ where $g(x;\omega)$ is the density of the politician's signal x in state ω . The politician thus weighs two considerations when deciding whether to pursue reform: her private assessment of climate risks and the electoral consequences of climate action; she chooses $a = 1$ when

$$\eta(x) + F(\mu(1)) \geq 1 - \eta(x) + F(\mu(0)).$$

Write $\Delta = F(\mu(1)) - F(\mu(0))$ as the difference in reelection probabilities, or the net electoral value of pursuing climate reforms. Then there is a cutoff signal $\tilde{x}^*$ at which the politician is indifferent between enacting climate policy and not, solving

$$\underbrace{2\eta(\tilde{x}^*) - 1}_{\text{net belief } a=1 \text{ appropriate}} + \underbrace{\Delta(\tilde{x}^*)}_{\text{net electoral return}} = 0.$$

The politician therefore pursues climate reform only when sufficiently convinced that climate risks justify action given expected electoral consequences. Because reelection incentives depend on voter expectations, the politician is more likely to adopt whichever policy the voter views as appropriate *ex ante*: she panders.⁹ In equilibrium, the probability of

⁹While pandering is well understood from canonical accountability models (e.g., [Canes-Wrone, Herron and Shotts 2001](#); [Maskin and Tirole 2004](#)), it arises in this model for a different reason. Here, pandering occurs not from selection incentives arising from incompetent or incongruent types, but because there is common value uncertainty.

climate action can be written as

$$A(\tilde{x}^*) = \pi \left(1 - G(\tilde{x}^*; 1)\right) + (1 - \pi) \left(1 - G(\tilde{x}^*; 0)\right).$$

If the politician knew the state of the world perfectly, she would engage in climate reforms only when appropriate, which occurs with probability π . The presence of pandering can therefore be diagnosed by comparing the equilibrium probability of action $A(\tilde{x}^*)$ to the prior: when $A(\tilde{x}^*) > \pi$, the politician enacts reforms more often than expected, pandering toward action; when $A(\tilde{x}^*) < \pi$, she acts less often, pandering toward inaction.

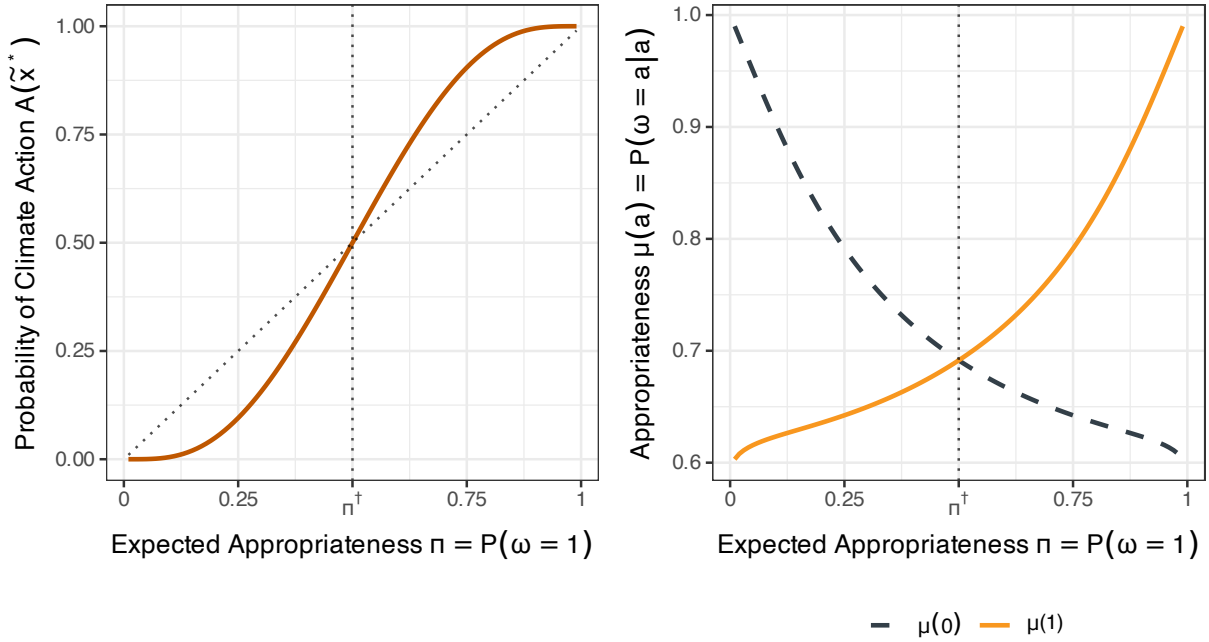


Figure 1: Pandering Equilibrium: Probability of Climate Action and Voter Beliefs
 $G(x; \omega) = N(\omega, 1)$, $F(\varepsilon) = N(0, 1)$

The effects of pandering on politician behavior are shown in the left panel of Figure 1. The solid line illustrates the equilibrium probability of climate action, which is increasing in π . Notably, it is below the 45° line when π is low, and above it when π is high. This disjuncture occurs because when expectations about climate severity are low, the voter perceives climate

reforms to be less appropriate, which demotivates the politician from pursuing them. The politician tempers climate reforms precisely because she knows that the voter would not perceive them to be appropriate. By contrast, when expectations about climate severity are high, the politician engages in climate reforms more than expected because the voter perceives it as the appropriate policy.

Voter beliefs about the politician's appropriateness are shown in the right panel of Figure 1. The solid line illustrates that $\mu(1)$, the belief that climate action is appropriate, increases as expectations of climate severity increase; the dashed line $\mu(0)$, the belief that climate inaction is appropriate, conversely decreases in expected severity. These curves intersect at a unique threshold $\pi^\dagger$, which is where the voter is indifferent between, and thus most uncertain about, climate policy appropriateness. As voter beliefs about appropriateness directly translate into electoral rewards, it is clear that the voter is more likely to penalize climate action when $\pi < \pi^\dagger$, and more likely to reward it when $\pi > \pi^\dagger$. This dynamic is central to what follows, as it is precisely this electoral reward that the interest group will seek to nullify through its choice of messaging strategy.

Effects of Special Interest Messaging

Building off of the preceding discussion, I now introduce the interest group that can design a public-facing report to minimize the probability that the politician enacts climate reforms. This is done indirectly by *persuading* voters to punish reforming politicians, because the message forms a lens through which the voter perceives climate policy appropriateness. Beyond the foundations from Proposition 1, Proposition 2 describes the additional features of the equilibrium that emerge once the interest group is introduced.

Proposition 2 *Let $\mathcal{I} = \{a_i, s\}$. There exists a unique equilibrium in which:*

- *as $\beta \rightarrow 1$, the equilibrium converges to the game without the interest group;*

- the probability of climate action is inverse U-shaped in β ;
- the special interest's optimal misreporting strategy is

$$\beta^*(\pi) = \begin{cases} 1 & \pi < \pi^\dagger \\ 0 & \pi \geq \pi^\dagger. \end{cases}$$

With the special interest group present, the voter now conditions his assessment of appropriateness, and consequently his reelection decision, on both the politician's action and the interest group's message: $\mu(a, s) = P(\omega = a | a, s)$. The politician is reelected with probability $F(\mu(a, s))$.

The interest group's messaging strategy changes the politician's electoral problem because she now anticipates how different public messages will shape voter reactions to climate policy. The politician weighs both her private assessment of climate risks and the electoral consequences associated with taking action given what the voter may know about climate severity. Let $\Delta(s) = F(\mu(1, s)) - F(\mu(0, s))$ denote the electoral return to climate action conditional on message s . The cutoff $\tilde{x}^*$ at which the politician is indifferent between climate action and inaction then solves

$$\underbrace{2\eta(\tilde{x}^*) - 1}_{\text{net belief } a=1 \text{ appropriate}} + \underbrace{(1 - \beta)\eta(\tilde{x}^*)\Delta(1)}_{\text{net electoral return if } s=1} + \underbrace{(1 - \eta(\tilde{x}^*) + \beta\eta(\tilde{x}^*))\Delta(0; \tilde{x}^*)}_{\text{net electoral return if } s=0} = 0.$$

Relative to the pandering baseline, the electoral return to climate action depends on how the voter would react under each possible message realization, weighted by the politician's belief about the likelihood of each message being observed. Importantly, the interest group's message functions as a lens through which the voter interprets the politician's policy choice. Because the group is biased toward the status quo, this effect is asymmetric: the message $s = 1$ is sent only when climate reform is truly appropriate, so observing it reveals severe

climate risks with certainty. By contrast, $s = 0$ is ambiguous: it may reflect truly limited climate risks or strategic misreporting when reforms are actually warranted.

The parameter β determines how often this ambiguity arises, and consequently how informative the message is overall. When $\beta = 0$, messaging is unbiased: the voter can perfectly infer whether climate reform is warranted, and actions inconsistent with the message are revealed to be inappropriate. But if $\beta = 1$, the group always misreports when climate reforms are warranted: the message carries no information about climate risks, and the voter rationally discounts it, reducing the game to the pandering baseline of Proposition 1. Between these extremes, the message is partially informative, and the voter combines it with the politician's action when assessing appropriateness.

Changes in β therefore affect climate policymaking through two competing effects. Greater misreporting makes anti-climate messages more common, which mechanically discourages reform by increasing the likelihood that voters observe messages downplaying climate risks. At the same time, such messages become less informative, causing the voter to discount them more heavily when evaluating the politician. These two effects generate a nonmonotonic relationship between misreporting and climate action: when misreporting is low, the marginal change on voter beliefs dominates and the politician's threshold falls, but when misreporting is widespread, voters already heavily discount the message and the mechanical effect dominates, raising the politician's threshold for reform. As a result, the equilibrium probability of climate action is inverse U-shaped in β .

Accordingly, for a special interest group seeking to deter climate action, the optimal choice of β is either to misreport entirely, $\beta = 1$, or to be an unbiased arbiter of the need for climate reform, $\beta = 0$.¹⁰ As shown in Proposition 2, the equilibrium level of misreporting

¹⁰This result departs from the canonical Bayesian persuasion framework (Kamenica and Gentzkow 2011; Alonso and Câmara 2016), in which optimal experiments are typically partially informative, occasionally triggering the sender's preferred action without fully revealing the state. The departure stems from the indirect structure of persuasion in this model: the special interest persuades the voter, who disciplines the politician through the electoral channel, rather than the politician herself.

depends on a connection between the prior π and the pandering threshold $\pi^\dagger$, with $\beta^* = 1$ when $\pi < \pi^\dagger$ and $\beta^* = 0$ when $\pi > \pi^\dagger$.

To see why, compare politician behavior under each case. If the interest group never misreports, $\beta = 0$, the voter can perfectly infer whether climate reform is warranted from the observed message. On the politician's side, electoral distortions disappear, and she pursues reform whenever she privately believes climate action is more likely than not to be appropriate. By contrast, if the interest group maximally misreports, choosing $\beta = 1$, its message becomes entirely uninformative and the game collapses to the pandering baseline from Proposition 1. Pandering incentives push the politician toward whichever policy the voter already expects to be appropriate *ex ante*.

The connection between persuasion and pandering therefore depends on voter expectations about climate severity. When expected climate risks are low, $\pi < \pi^\dagger$, the voter is likely to penalize climate reform relative to inaction. The politician consequently panders toward the status quo by pursuing reform less frequently than underlying climate conditions warrant. Under these conditions, anti-climate interests benefit from maximal misreporting because uninformative messaging preserves these electoral incentives against climate action. Even though the voter rationally discounts the message, the politician is still deterred because taking climate action remains electorally risky.

However, when expected climate severity is high, $\pi > \pi^\dagger$, the politician's incentive to pander flips toward implementing climate action. The special interest group can counter, and entirely neutralize, this electoral gain by sending a truthful, unbiased message about the need for climate reforms. Counterintuitively, by making the message $s = 0$ a reliable signal of $\omega = 0$, the group equips the voter with information to punish a politician who pursues climate reform when it is not appropriate. This electoral threat shuts down the politician's pandering pressure toward action, making her more conservative in her propensity to implement climate reforms. When *ex ante* climate severity is high, a special interest group

thus optimally persuades politicians away from climate action with the threat of electoral removal by a perfectly informed voter.

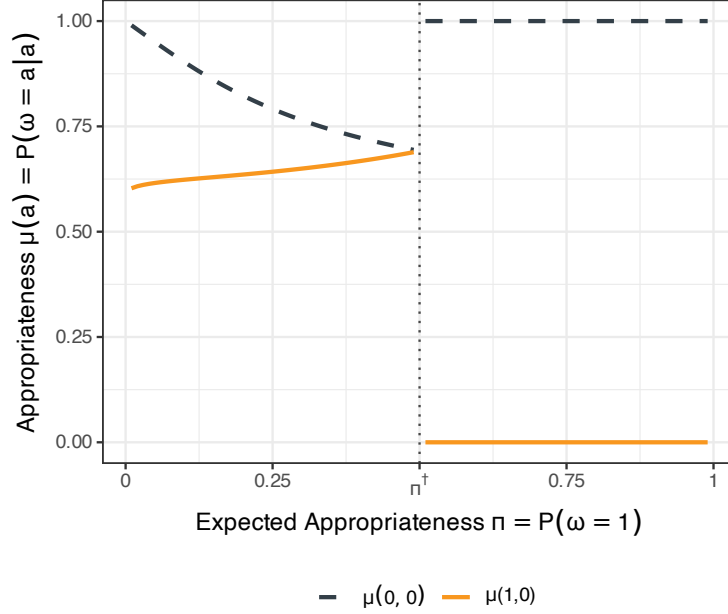


Figure 2: Persuasion Equilibrium: Voter Beliefs
 $G(x; \omega) = N(\omega, 1)$, $F(\varepsilon) = N(0, 1)$

Figure 2 illustrates the equilibrium consequences of the special interest's strategy for voter inference. The solid line plots $\mu(1, 0; \tilde{x}^*)$, the voter's posterior that climate action is appropriate when $s = 0$, and the dashed line depicts $\mu(0, 0; \tilde{x}^*)$, the posterior that inaction is warranted when $s = 0$. To the left of $\pi^\dagger$, where the interest group fully misreports, the voter rationally discounts the group's message as uninformative and falls back on the prior to evaluate the politician's policy choice. The posteriors mirror the pandering baseline of Proposition 1: the voter becomes more confident that climate action is appropriate as expected severity rises. However, there is a sharp discontinuity at $\pi = \pi^\dagger$, when the special interest shifts from maximal misreporting to reporting an unbiased signal about the need for climate reforms. To the right of $\pi^\dagger$, the message $s = 0$ is perfectly informative of $\omega = 0$, and voter inference collapses to certainty: $\mu(0, 0; \tilde{x}^*) = 1$ and $\mu(1, 0; \tilde{x}^*) = 0$ for any value of the

prior. The interest group thus conditions what the voter can learn about climate risks and how they interpret the policy choices they observe.

Effects of International Climate Action

I now demonstrate that foreign climate policies shape domestic politics by affecting how voters evaluate whether climate reforms are warranted. Proposition 3 characterizes the equilibrium with *international cooperation* and establishes additional core results. Notably, foreign climate actions generate informational spillovers that link domestic political incentives across countries, which can produce either strategic complementarity or substitutability in climate policymaking. Moreover, the political importance of foreign climate action depends critically on the information disseminated by special interests.

Proposition 3 *Let $\mathcal{I} = \{a_i, s, a_j\}$. There exists a unique equilibrium in which:*

- *international climate actions can either be strategic complements or substitutes;*
- *observation of international climate policies increases climate action relative to the domestic model when $\pi < \pi^\dagger$ and has no effect when $\pi > \pi^\dagger$;*
- *bilateral climate action increases in π , bilateral inaction decreases in π , and unilateral climate action is inverse-U shaped in π , peaking at $\pi^\dagger$.*

Domestic politicians now anticipate that voters will evaluate climate policy in light of foreign actions as well. When foreign governments pursue climate reforms, voters may interpret those actions as evidence that climate risks are severe and that reform is warranted. Foreign climate policy therefore becomes an informational cue that shapes domestic electoral incentives. The voter in country i observes the triple (a_i, s, a_j) and forms posterior

belief $\mu_i(a_i, s, a_j)$ about whether domestic climate action is appropriate. Politicians consequently care not only about how voters interpret domestic policy choices and special interest messaging, but also about how voters interpret climate action from abroad.

Incorporating the foreign action into electoral incentives, write $\Delta_i(s, a_j) = F(\mu_i(1, s, a_j)) - F(\mu_i(0, s, a_j))$ as the electoral return to climate action conditional on the special interest's message and foreign policy realization. Let $p_{j\omega} = P(a_j = 1|\omega)$ be politician i 's belief about country j 's likelihood of pursuing climate reforms in state ω . The unique symmetric equilibrium cutoff $\tilde{x}^*$ satisfies

$$\begin{aligned}
& \underbrace{2\eta(\tilde{x}^*) - 1}_{\text{net belief } a_i=1 \text{ appropriate}} + \underbrace{(1 - \beta)\eta(\tilde{x}^*)\Delta_i(1, a_j)}_{\text{net electoral return if } s=1} \\
& + \underbrace{\left(\beta\eta(\tilde{x}^*)p_{j1}(\tilde{x}^*) + (1 - \eta(\tilde{x}^*))p_{j0}(\tilde{x}^*)\right)\Delta_i(0, 1; \tilde{x}^*)}_{\text{net electoral return if } s=0 \text{ and } a_j=1} \\
& + \underbrace{\left(\beta\eta(\tilde{x}^*)(1 - p_{j1}(\tilde{x}^*)) + (1 - \eta(\tilde{x}^*))(1 - p_{j0}(\tilde{x}^*))\right)\Delta_i(0, 0; \tilde{x}^*)}_{\text{net electoral return if } s=0 \text{ and } a_j=0} = 0.
\end{aligned}$$

The politician's willingness to pursue climate reform now depends not only on her own assessment of climate risks, but also on how voters interpret her actions in light of both special interest messaging and foreign climate policy. Note that foreign climate policy adoption does not affect the politician's preferences for action independently of domestic concerns: there are no direct free-rider or coordination effects. Instead, any international effects on domestic climate policy adoption operate through how the voter interprets the appropriateness of climate action given foreign behavior.

Through domestic politics, international climate action shapes domestic reform incentives in two ways. The first channel operates through politicians' expectations about foreign behavior. If the foreign politician becomes more likely to pursue climate reforms, the domestic politician anticipates more instances in which her own action will coincide with foreign

policies. Voters, seeking to infer whether climate reform is warranted, therefore reward domestic action more when foreign governments also adopt climate policies—in equilibrium, $\Delta_i(0, 1; \tilde{x}^*) \geq \Delta_i(0, 0; \tilde{x}^*)$. Foreign climate action reassures voters that reform is appropriate, increasing the domestic politician’s electoral incentive to act. Through this channel, climate policies behave as strategic complements: a greater expected willingness to act abroad strengthens incentives for domestic climate reform.

The second force operates through voter inference. If the foreign politician adopts climate reforms even when signals of climate severity are weak, then foreign action provides only limited evidence that reform is truly warranted. By contrast, when the foreign politician acts selectively, pursuing reform only after sufficiently strong signals, foreign climate action becomes a more informative signal of underlying climate risks. Domestic voters then reward climate action more strongly when it coincides with selective foreign reforms, increasing the domestic politician’s electoral incentive to act—formally, $\Delta_i(0, a_j)$ is increasing in $\tilde{x}^*$. Through this channel, climate policies act as strategic substitutes: greater selectivity abroad reduces the frequency of foreign action while increasing its informational value domestically, so the two countries’ propensities to reform move in opposite directions.

Hence, because these complementary and substitutable forces are both present, the overall strategic relationship between countries depends on which effect is stronger in equilibrium.¹¹ Existing theories of international climate politics typically impose a strategic structure by assumption—substitutability when the focus is on free-riding incentives (e.g., [Kennard and Schnakenberg 2023](#)), or complementarity when the focus is on institutional coordination (e.g., [Barrett 2016](#))—and then derive implications conditional on that structure. Here, by contrast, the strategic relationship between countries emerges endogenously from how voters interpret foreign climate actions.

¹¹It should be noted that, formally, these strategic effects are quantitatively smaller than the domestic politician’s response to her own signal x_i , thereby guaranteeing a unique equilibrium (see Lemma A.8).

A second consequence of the interconnectedness of international and domestic politics is that the political importance of foreign climate action is itself conditional on what voters can learn about climate severity. The special interest’s optimal strategy is unchanged from Proposition 2: $\beta^* = 1$ when $\pi < \pi^\dagger$ and $\beta^* = 0$ when $\pi > \pi^\dagger$. But this messaging strategy disciplines what voters can learn from foreign action. When $\pi < \pi^\dagger$ and special interests maximally misreport, voters lack a reliable domestic signal of climate policy appropriateness, so foreign climate action becomes a meaningful alternative source of information. The domestic politician anticipates this, and her electoral incentive to pursue reform rises when foreign action is expected. The likelihood she takes action is therefore greater than in a model without international cooperation. But when $\pi > \pi^\dagger$ and special interests provide unbiased messaging, foreign action provides no additional informational content. International political dynamics therefore matter most for domestic climate action precisely when domestic informational conditions are least conducive to reform, and their effect attenuates when special interests send unbiased messages about climate severity.

The preceding analysis characterizes climate policy choices at the country level. Aggregating these choices across countries yields implications for international climate outcomes, and three outcomes are possible. Recall that the probability of climate action in either country is $A(\tilde{x}^*)$, so we have: bilateral climate action, in which both countries pursue reforms, occurring with probability $A(\tilde{x}^*)^2$; bilateral climate inaction, in which neither pursues reforms, occurring with probability $(1 - A(\tilde{x}^*))^2$; and unilateral climate action, in which exactly one country acts, occurring with probability $2A(\tilde{x}^*)(1 - A(\tilde{x}^*))$. Extant theories typically focus on one of these outcomes. International accounts emphasize the tragedy of the commons to explain mutual inaction or the institutional structures sustaining conditional cooperation leading to bilateral action (Barrett 2003; Rowan 2025), while domestic accounts focus on unilateral action (Aklin and Mildenberger 2020; Ross 2025). This model instead provides a unified framework: all three outcomes occur with positive probability, with their relative

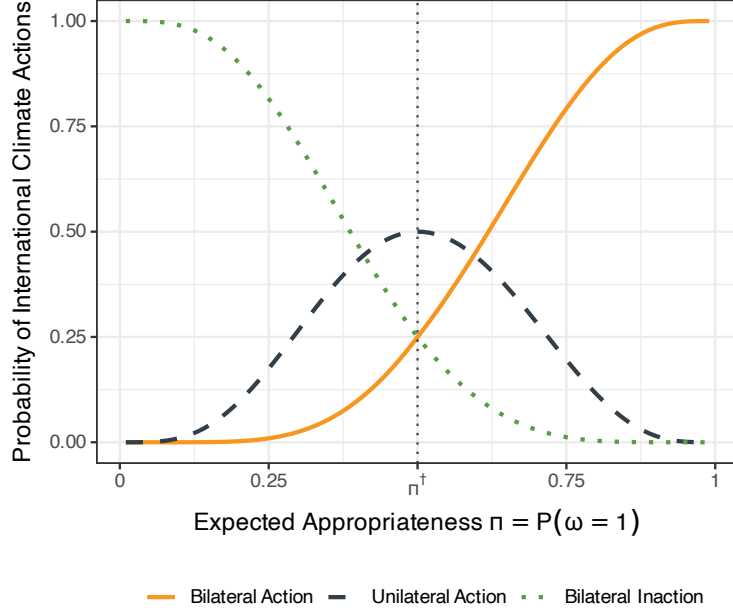


Figure 3: Cooperation Equilibrium: Probability of International Climate Action
 $G(x; \omega) = N(\omega, 1)$, $F(\varepsilon) = N(0, 1)$

likelihoods determined by the same informational mechanisms that shape domestic policy-making.

Figure 3 illustrates how these international climate policy outcomes evolve as a function of expected climate severity. The solid line plots the probability of bilateral action; the dotted line plots the probability of bilateral inaction; and the dashed line plots the probability of unilateral action. Intuitively, bilateral action is increasing in π , and bilateral inaction is decreasing in π .¹² The probability of unilateral action, however, is inverse-U shaped in π and peaks at $\pi = \pi^\dagger$, where the variance of policy outcomes across countries is maximized. Substantively, this means that unilateral, or “uncoordinated,” outcomes are most likely when electoral returns to climate policy are most muted. Around $\pi = \pi^\dagger$, pandering incentives

¹²These curves are not symmetric because the special interest’s optimal misreporting shifts as expected severity crosses $\pi^\dagger$. Bilateral climate action rises in π but converges more slowly toward 1 in its upper tail because unbiased reporting diminishes pandering incentives: as π approaches 1, the special interest shifts to unbiased messaging, leading politicians to become more conservative in their pursuit of climate reforms, slowing the probability of action.

are weakest, and voters are most uncertain *ex ante* about the appropriateness of reforms. With electoral incentives muted, politicians rely more heavily on their private signals when determining policy. Because signals are independently realized across countries, governments may reach different conclusions about whether reform is warranted, increasing the likelihood of unilateral action. This demonstrates that the international system’s coordination failures need not be produced by free-riding incentives or the absence of cooperation institutions, but by the inferential structure of domestic politics in conditions where appropriateness is genuinely uncertain.

Empirical Implications

To assess the empirical plausibility of the theory’s implications, I present qualitative and quantitative evidence of the mechanisms at play. I first describe variation in special interest messaging behavior, focusing on ExxonMobil’s strategies to craft public reports to downplay climate risks when severity was low, ultimately shifting toward acknowledgment of climate harms in 2014. This evolution is consistent with the theory’s expectation that misreporting becomes less politically sustainable as public concern about climate risks rises. Quantitatively, I examine descriptive patterns in cross-national survey data on voter beliefs about climate policy and climate law adoption.

Trajectories of Exxon’s Public Messaging

A substantial share of historical greenhouse gas emissions can be traced to a relatively small number of fossil fuel firms, many of which had strong incentives to oppose climate regulation (Heede 2014; Ekwurzel et al. 2017). Existing research documents that these firms frequently emphasized uncertainty surrounding climate science in public communications, seeking to weaken support for environmental regulation (Oreskes and Conway 2011; Brulle

2014; Williams et al. 2022). Over time, however, several major firms shifted toward greater public acknowledgment of climate risks and limited support for climate policy.

ExxonMobil offers the clearest documented illustration of this trajectory, and the regulatory and academic attention paid to the company makes its messaging history especially well-traced. Figure 4 displays a timeline of some key events pertaining to Exxon’s disclosure of climate-related information (see Appendix B for further details on sources in the figure). Beginning in the late 1990s and through the mid-2000s, Exxon exploited the uncertainty inherent to climate science, engaging in public communication campaigns that emphasized uncertainty surrounding climate risks, attempting to convince citizens that climate change did not warrant broad policy action. While climate reforms would run counter to Exxon’s interests, the company nevertheless shifted its messaging in 2014, when it publicly acknowledged its role in fostering climate risks. Since then, Exxon has advocated for policy solutions like carbon pricing that both recognize the climate threat and take steps toward addressing it. This pattern of accepting climate science and policies, even lobbying in favor of reforms, has been documented within extant literature (Kennard 2020; Green et al. 2022).

Exxon’s public messaging campaign dovetailed with a global movement to delay climate action by downplaying risks to audiences across borders (Schlichting 2013). While much of Exxon’s documented public-facing communication targeted American audiences directly, these messages circulated internationally through transnational media coverage, and the company directly funded a network of European think tanks. More broadly, Exxon was hardly unique: Shell, BP, and other firms associated with the Global Climate Coalition similarly produced internal assessments acknowledging the scientific basis of anthropogenic climate change while publicly emphasizing uncertainty surrounding its consequences. The historical record therefore points to a broader landscape in which major fossil fuel interests consistently cast doubt on the severity of climate risks, even beyond Exxon’s own messaging

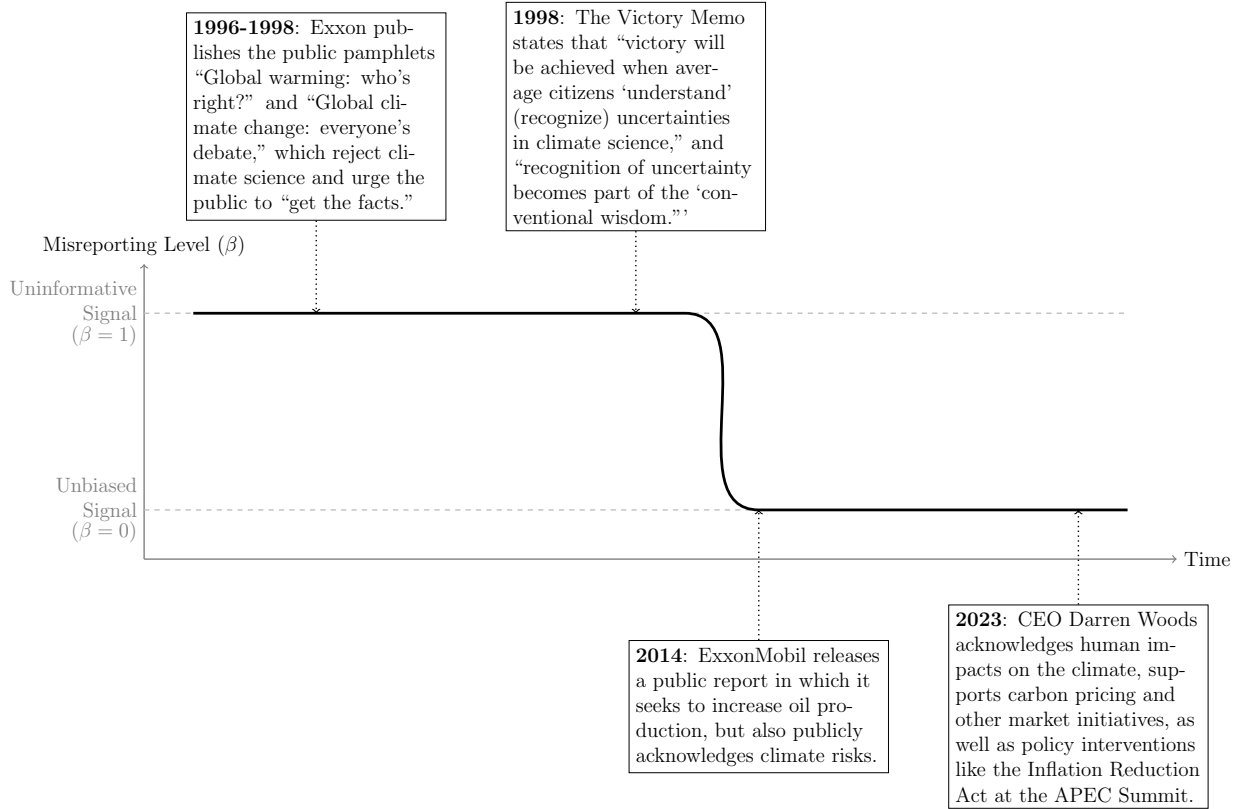


Figure 4: Variation in Exxon’s Climate Messaging

efforts.¹³

Exxon’s shift in public communications provides a particularly visible and well-documented marker of a broader change in the information available to the public about the severity of climate change. By the mid-2010s, many of the most prominent firms that had historically emphasized uncertainty surrounding climate risks had moved toward public acknowledgment of climate harms and, in some cases, support for limited climate policy interventions (Green et al. 2022). In the quantitative analysis that follows, I leverage Exxon’s 2014 acknowledgment of climate risks as an empirical proxy for a wider change in the signals available to citizens about climate severity, as the company’s communications reflected a broader trans-

¹³For example, BP’s carbon footprint calculator, launched in 2004, reframed climate responsibility around individual consumption choices rather than fossil fuel production, potentially dampening support for broad regulatory responses.

formation in the public discourse surrounding climate change. Interpreted in this way, the 2014 pivot provides a useful empirical analogue to the model’s shift in optimal misreporting at $\pi = \pi^\dagger$.

Voter Beliefs and Climate Law Adoption

The theory generates observable implications for how individuals evaluate climate policy appropriateness as a function of three factors: their own government’s climate policies, special interest messaging about climate risks, and climate policies adopted abroad. In particular, the model implies that beliefs about whether climate action is warranted should vary systematically across the eight possible cases generated by permuting these three inputs. The key empirical implications are a set of directional comparisons—relationships between beliefs given variation in these inputs—in how citizens interpret climate policy choices. I assess these implications using cross-national public opinion data and data on climate law adoption.

It should be noted upfront that the following exercise should be viewed as a probe into the plausibility of the theory based on the set of directional comparisons, not a causal test of the theory’s propositions. The theory provides a mechanism through which domestic adoption, foreign adoption, or special interest messaging each exert an equilibrium effect on beliefs, which can be difficult to parse. Notwithstanding, it does imply that beliefs about climate policy appropriateness should be directionally ordered based on the realized values of the three inputs across the eight cases. The empirical exercise thus examines whether the observed, empirical directional comparisons match theoretical expectations.

Data. To measure individual beliefs about climate policy appropriateness, I use repeated cross-sectional data from Eurobarometer to capture public opinion on the seriousness of climate change across 27 European countries and seven survey waves between 2011 and

2023. Eurobarometer asks citizens to assess whether they believe that climate change is a serious problem on a scale from 1 to 10, where 1 is “not at all a serious problem” and 10 is “an extremely serious problem.”¹⁴

To measure climate action, I compile a list of mitigation-related laws adopted by each country in a given year from the Climate Change Laws of the World dataset (Nachmany et al. 2017). I create a binary indicator for whether a country has adopted a climate law in a given year, paralleling the model’s conception of climate action. Since climate law adoption is measured at the country-year level, I aggregate respondents’ beliefs about the seriousness of climate change by taking the average within countries for each surveyed year.

I leverage Exxon’s 2014 public acknowledgment of climate harms as an observable marker of a broader shift in special interest messaging about the severity of climate change. Because Exxon was among the most visible firms involved in public debates over climate policy, and because similar rhetorical shifts occurred across a wider set of fossil fuel interests during this period, its 2014 pivot provides a useful proxy for changes in the information available to citizens. I thus interpret time prior to 2014 as representing the case where $\pi < \pi^\dagger$, in which the optimal messaging strategy is full misreporting whereby $s = 0$ is always realized. Time after 2014 represents when $\pi > \pi^\dagger$, in particular following a realization of the signal $s = 1$ given shifts in optimal reporting.

Because the theory concerns how domestic and foreign climate policies jointly shape public beliefs, I structure the data as a dyadic panel linking each country to every other European country in a given year. For each dyad-year, I observe whether the home European country adopted a climate law, whether the foreign partner country adopted a climate law, and whether the observation occurs before or after Exxon’s 2014 messaging shift. I then compare average beliefs within EU countries (demeaned by dyad and year) about climate seriousness

¹⁴While perceived seriousness is not identical to the model’s concept of climate policy appropriateness, greater concern about climate risks should closely track beliefs that stronger climate policies are warranted. Given data availability this is a reasonable proxy for the model’s core concept.

across these combinations. To reduce concerns about simultaneity, climate law adoption is lagged by one year. Confidence intervals are obtained from dyad-clustered standard errors. Additional details are provided in Appendix C.

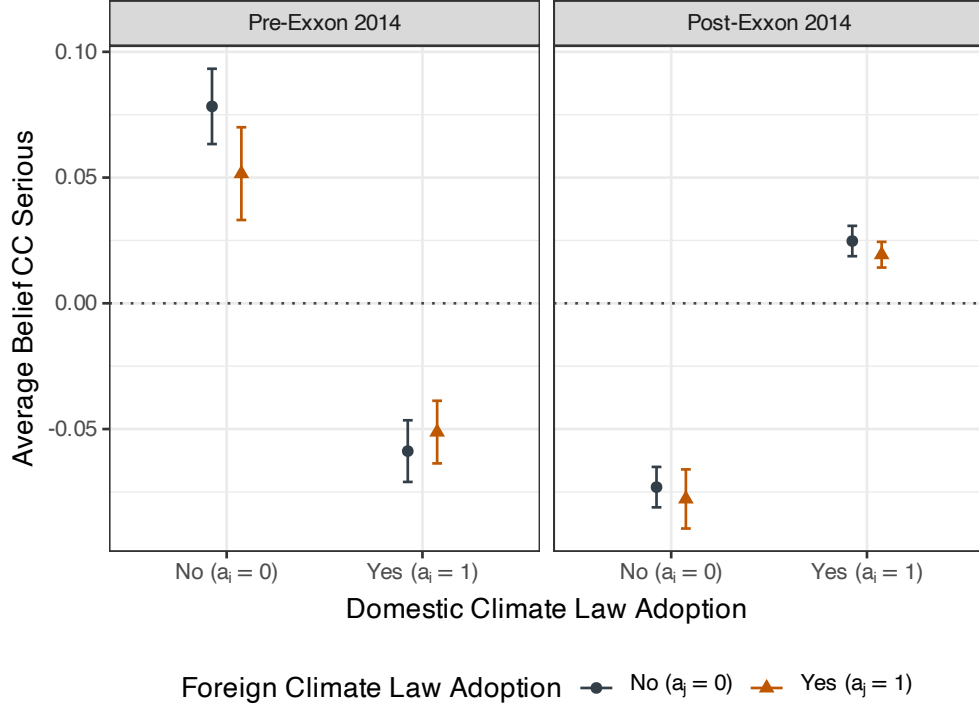


Figure 5: Average Beliefs, Exxon Messaging, and Climate Law Adoption
Note: Average country-level beliefs that climate change is a serious problem, demeaned by dyad and year, with 95% confidence intervals and dyad-clustered robust standard errors.

Results. Figure 5 reports average beliefs that climate change is a serious problem from the vantage point of citizens in country i in year t , $E[\text{CC Serious}|a_i, s, a_j]$, given domestic and foreign climate law adoption in the previous year and expected special interest messaging (pre- or post-2014). These eight means correspond to the eight cases generated by the model and provide empirical analogues to the voter posteriors $\mu_i(a_i, s, a_j)$. The theory implies an ordering of beliefs across these cases, or expectations about the sign of the differences of these means. It is these comparisons with which I evaluate the plausibility of the theory.

Given the eight posterior beliefs, there are twelve possible differences; see Remark A.1 for

a formal exposition, though they are simply the differences in equilibrium posterior beliefs implied by Proposition 3. The model signs eleven of them cleanly and, as I discuss below, leaves one theoretically ambiguous. Figure 6 reports these comparisons, obtained by varying each input while holding the other two fixed. The contrasts are grouped according to whether they isolate the effects of domestic climate policy adoption, special interest messaging, or foreign climate policy adoption.

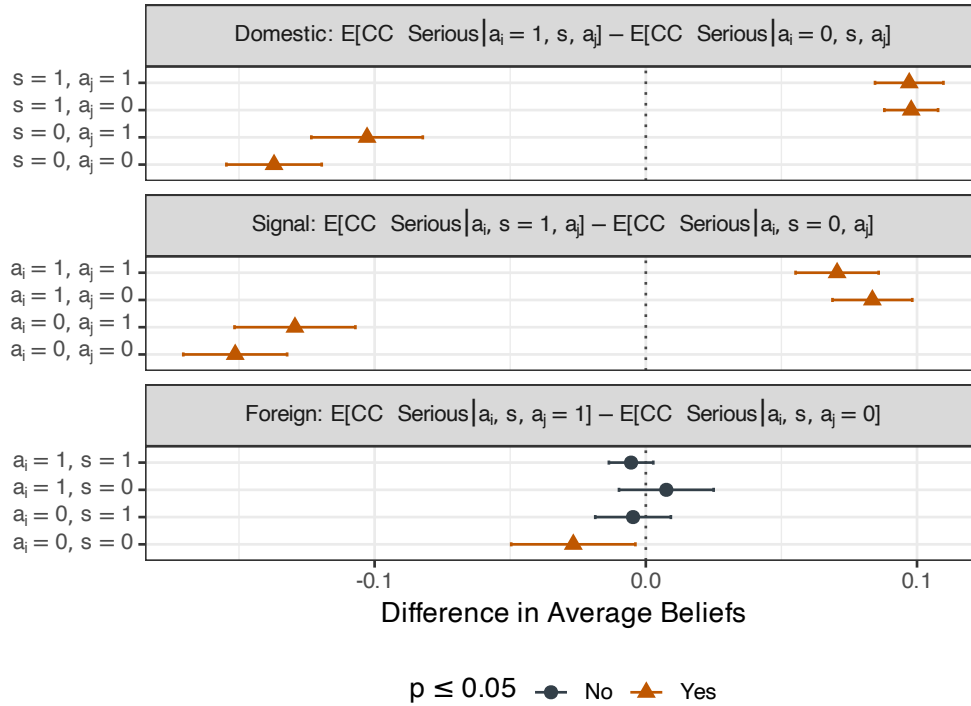


Figure 6: Differences in Average Beliefs

Note: Differences in average country-level beliefs that climate change is a serious problem, demeaned by dyad and year, with 95% confidence intervals and dyad-clustered robust standard errors.

The top panel varies domestic climate policy adoption. When Exxon's messaging indicates that climate action is warranted, domestic climate law adoption is consistently associated with higher perceived climate seriousness compared to domestic inaction, regardless of whether a foreign country also adopts a climate law or not (0.971 points and 0.979 points respectively). When Exxon's messaging instead suggests that climate action is unnecessary, the

theory implies that beliefs should generally favor whichever domestic policy is coordinated with the foreign country’s behavior. Consistent with this expectation, when neither country adopts a climate law, domestic action is associated with lower perceived climate seriousness than domestic inaction (-0.137 points). In this setting, coordinated inaction is viewed more favorably than a unilateral departure toward climate action. By the same logic, the theory anticipates that when the foreign country adopts a climate law, coordinated climate action should be viewed more favorably than domestic inaction. This contrast, however, is the one whose sign the theory does not pin down: conditional on foreign climate action, domestic climate law adoption is associated with lower perceived climate seriousness than domestic inaction (-0.103 points).¹⁵ Thus, the domestic comparisons are broadly consistent with the model’s informational logic.

The middle panel varies the public signal from Exxon while holding domestic and foreign climate policies fixed. These contrasts directly capture the role of special interest messaging in the model. Across all four comparisons, beliefs about climate seriousness move in the direction implied by the signal. When the domestic country does not adopt a climate law, perceived climate seriousness is lower after Exxon’s public acknowledgment of climate harms than before (-0.129 points when a foreign country adopts a law, -0.151 points when it does not), consistent with the idea that inaction is viewed less favorably when public signals indicate greater climate risk. Conversely, when climate laws are adopted at home, perceived seriousness is higher following Exxon’s messaging shift (0.071 points when a foreign country adopts a law, 0.084 points when it does not).

The bottom panel examines informational spillovers from foreign climate action. When Exxon’s messaging suggests that climate action is unnecessary, foreign climate action is

¹⁵As demonstrated in Remark A.1, the sign of this relationship ($\mu_i(1, 0, 1) - \mu_i(0, 0, 1)$) is theoretically ambiguous: there is a second threshold $\pi^{\dagger\dagger} < \pi^{\dagger}$ below which this difference is negative and above which this difference is positive. The empirical design cannot cleanly separate the prior π between the relevant theoretical thresholds, namely $\pi < \pi^{\dagger\dagger}$ versus $\pi^{\dagger\dagger} < \pi < \pi^{\dagger}$.

associated with meaningful shifts in beliefs about climate seriousness. The largest effect occurs when the home country does not adopt a climate law, and foreign climate action is associated with significantly lower perceived climate seriousness compared to foreign inaction (-0.027 points). When the home country does adopt a climate law, the estimated difference is directionally consistent with the theory's expectation that coordinated climate action should be viewed more favorably than mismatched policy choices, though the estimate is not statistically distinguishable from zero (0.008 points). By contrast, once Exxon's messaging indicates that climate action is warranted, the estimated effects of foreign climate law adoption are small and attenuate toward zero. While these null estimates do not by themselves demonstrate the absence of foreign policy effects, they are consistent with the broader pattern anticipated by the theory: foreign climate actions appear most informative when special interest messaging about climate severity is weakest.

The most distinctive implication of the theory concerns the conditional informational value of foreign climate action. Prior to the shift in public messaging, foreign climate policies are associated with meaningful changes in climate beliefs. Following the shift, these associations attenuate sharply toward zero. This pattern is consistent with the theory's claim that foreign actions serve as a substitute source of information when direct information about climate severity is limited, but become less informative once citizens can assess climate risks more directly.

Alternative explanations. While I again emphasize that this empirical exercise serves to assess the plausibility of the theory's core informational mechanism rather than as a causal test, the analysis is useful because it speaks to the theory's expectations about the difference in posterior beliefs about climate policy appropriateness in several joint scenarios. It thus helps to rule out alternative explanations that may have expected one or several, but not all, of the above contrasts. One such alternative is a salience account, in which climate change

simply becomes more available and more readily judged as serious over time (e.g., [Egan and Mullin 2012](#); [Egan, Konisky and Mullin 2022](#)), and in which observed policy activity—at home or abroad—raises perceived seriousness by keeping the issue before the public.

Such an account can rationalize a general rise in beliefs after 2014, as well as a positive association between policy adoption and perceived seriousness. However, such an account would likely expect an increase in perceived seriousness across the board following policy adoption; what is observed instead is a conditional relationship between policy adoption and special interest messaging. Under uninformative messaging pre-2014, coordinated inaction is viewed more favorably than a unilateral departure toward action, so domestic adoption is associated with lower perceived seriousness, while under informative messaging post 2014, domestic adoption is associated with higher perceived seriousness. A mechanism that induces greater seriousness with policy adoption cannot deliver a contrast that flips sign with the informativeness of the signal, because policy adoption is present in both cases.

A related concern is that the Paris Agreement, negotiated in 2015 and roughly coincident with Exxon’s messaging shift, drives beliefs about climate seriousness rather than assessments of policy appropriateness. Such an account could explain a general rise in perceived seriousness around Exxon’s pivot, as the salience of international cooperation increased. However, if deepening international cooperation were the operative force, foreign climate action should become more consequential for domestic beliefs as cooperation intensifies after 2015. Instead, we fail to observe meaningful shifts in beliefs given changes in foreign action in recent years, which is consistent with the informational spillovers mechanism.

While there may be other alternative accounts in addition to issue salience that describe variation in climate beliefs, it is the joint set—and in particular the conditional dependence of foreign action’s informational value on special interest messaging—that is difficult to reproduce without a mechanism in which voters use the actions of others as evidence about whether domestic climate policy is warranted. While other mechanisms may also generate

updating in response to foreign climate policies, they must additionally account for why the relationship between foreign policy adoption and climate beliefs weakens precisely when citizens are exposed to more informative signals about climate risks.

More generally, a limitation of the analysis is that dating the shift in special interest messaging to a single year does not on its own isolate a change in π —and thus a change in equilibrium reporting—as opposed to other developments concentrated in the mid-2010s, including declining costs of renewable generation and secular trends in climate concern. De-meaning by dyad and year absorbs shocks common to a given year or a particular dyad of countries. The interaction of the post-2014 messaging shift indicator with policy adoption provides within-period variation but cannot separate the messaging channel from these potential coincident shifts. The inferential weight of the analysis accordingly rests on the conditional structure of the ordering—the sign reversal of domestic adoption across messaging regimes and the attenuation of foreign action’s informational value once messaging is informative—which a common secular trend does not explain.

Conclusion

A central debate in the climate politics literature concerns whether climate policy outcomes are best understood as products of domestic distributive conflict ([Aklin and Mildenberger 2020](#); [Ross 2025](#); [Gazmararian and Tingley 2026](#))—with international politics mattering little ([Urpelainen and Van de Graaf 2018](#))—or of strategic interdependence among states ([Barrett 2003; 2016](#); [Kennard and Schnakenberg 2023](#); [Rowan 2025](#)). The theory developed here suggests these perspectives are not in tension. The model is fundamentally domestic in its political logic: politicians pursue climate policy with an eye toward reelection, and there is distributive conflict between voters and a special interest that influences policy indirectly through public messaging. Yet international forces still matter because voters use climate

policies adopted abroad to evaluate whether domestic climate reforms are warranted. International politics therefore affects domestic policymaking by shaping electoral assessments of policy appropriateness.

The empirical evidence is broadly consistent with the theory's central informational logic. Beliefs about climate seriousness vary across combinations of domestic climate action, foreign climate action, and special interest messaging in ways largely congruent with expectations about informational spillovers and climate policy appropriateness. In particular, the political significance of both domestic and foreign climate actions varies with the signals citizens receive about climate risks. Foreign climate actions are most strongly associated with domestic beliefs when special interest messaging provides little additional information about climate severity and become substantially less consequential once such signals are available. These patterns are difficult to reconcile with accounts in which international climate politics operates solely through free-riding incentives or direct coordination effects. Instead, they suggest that foreign climate actions shape domestic politics in part by serving as informational cues that citizens use to evaluate whether climate policy is warranted.

More broadly, the theory suggests that international climate politics retains independent political importance even if domestic distributive conflict may be central to policymaking. Foreign policy choices become opportunities to learn about domestic climate policy appropriateness, and climate action can propagate across countries even in the absence of formal coordination mechanisms or collective action concerns. Overlooking these informational dynamics risks misreading both the prospects for domestic climate reform and the political consequences of international climate action. As publics across countries continue to revise their assessments of climate risks, and as they increasingly draw on the climate actions of others in forming those assessments, whether and how climate policy advances will depend on the information available to voters and how it is interpreted, as much as on the distributive politics of decarbonization.

References

- Aklin, Michaël and Johannes Urpelainen. 2013. “Political Competition, Path Dependence, and the Strategy of Sustainable Energy Transitions.” *American Journal of Political Science* 57(3):643–658.
- Aklin, Michaël and Matto Mildemberger. 2020. “Prisoners of the Wrong Dilemma: Why Distributive Conflict, Not Collective Action, Characterizes the Politics of Climate Change.” *Global Environmental Politics* 20(4):4–27.
- Alonso, Ricardo and Odilon Câmara. 2016. “Persuading Voters.” *American Economic Review* 106(11):3590–3605.
- Balcazar, Carlos Felipe and Amanda Kennard. 2025. “Climate Change and Political Mobilization: Theory and Evidence from India.” <https://drive.google.com/file/d/14yiR15XTWylc8-xE3RosKL03i2-Ta989/view> .
- Barrett, Scott. 2003. *Environment and Statecraft: The Strategy of Environmental Treaty-Making*. Oxford: Oxford University Press.
- Barrett, Scott. 2016. “Coordination vs. voluntarism and enforcement in sustaining international environmental cooperation.” *Proceedings of the National Academy of Sciences* 113(51):14515–14522.
- Battaglini, Marco and Bård Harstad. 2020. “The Political Economy of Weak Treaties.” *Journal of Political Economy* 128(2):544–590.
- Bayer, Patrick and Federica Genovese. 2020. “Beliefs About Consequences from Climate Action Under Weak Climate Institutions: Sectors, Home Bias, and International Embeddedness.” *Global Environmental Politics* 20(4):28–50.

- Bechtel, Michael M. and Kenneth F. Scheve. 2013. “Mass support for global climate agreements depends on institutional design.” *Proceedings of the National Academy of Sciences* 110(34):13763–13768.
- Beiser-McGrath, Liam F. and Thomas Bernauer. 2019. “Commitment failures are unlikely to undermine public support for the Paris agreement.” *Nature Climate Change* 9(3):248–252.
- Bernauer, Thomas, Liang Dong, Liam F. McGrath, Irina Shaymerdenova and Haibin Zhang. 2016. “Unilateral or Reciprocal Climate Policy? Experimental Evidence from China | Article | Politics and Governance.”.
- Bernauer, Thomas and Robert Gampfer. 2015. “How robust is public support for unilateral climate policy?” *Environmental Science & Policy* 54:316–330.
- Bolet, Diane, Fergus Green and Mikel González-Eguino. 2024. “How to Get Coal Country to Vote for Climate Policy: The Effect of a “Just Transition Agreement” on Spanish Election Results.” *American Political Science Review* 118(3):1344 – 1359.
- Brulle, Robert J. 2014. “Institutionalizing delay: foundation funding and the creation of U.S. climate change counter-movement organizations.” *Climatic Change* 122(4):681–694.
- Brulle, Robert J. 2018. “The climate lobby: a sectoral analysis of lobbying spending on climate change in the USA, 2000 to 2016.” *Climatic Change* 149(3):289–303.
- Brulle, Robert J. 2021. “Networks of Opposition: A Structural Analysis of U.S. Climate Change Countermovement Coalitions 1989–2015.” *Sociological Inquiry* 91(3):603–624.
- Canes-Wrone, Brandice, Michael C. Herron and Kenneth W. Shotts. 2001. “Leadership and Pandering: A Theory of Executive Policymaking.” *American Journal of Political Science* 45(3):532–550.

- Cheon, Andrew and Johannes Urpelainen. 2013. “How do Competing Interest Groups Influence Environmental Policy? The Case of Renewable Electricity in Industrialized Democracies, 1989–2007.” *Political Studies* 61(4):874–897.
- Colantone, Italo, Livio Di Lonardo, Yotam Margalit and Marco Percoco. 2024. “The Political Consequences of Green Policies: Evidence from Italy.” *American Political Science Review* 118(1):108–126.
- Colgan, Jeff D., Jessica F. Green and Thomas N. Hale. 2021. “Asset Revaluation and the Existential Politics of Climate Change.” *International Organization* 75(2):586–610.
- Cory, Jared, Michael Lerner and Iain Osgood. 2021. “Supply Chain Linkages and the Extended Carbon Coalition.” *American Journal of Political Science* 65(1):69–87.
- Druckman, James N. and Mary C. McGrath. 2019. “The evidence for motivated reasoning in climate change preference formation.” *Nature Climate Change* 9(2):111–119.
- Dunlap, Riley and A.M. McCright. 2011. Organized Climate Change Denial. In *The Oxford Handbook of Climate Change and Society*, ed. John S. Dryzek, Richard B. Norgaard and David Schlosberg. pp. 144–160.
- Dunlap, Riley E. and Aaron M. McCright. 2015. Challenging Climate Change: The Denial Countermovement. In *Climate Change and Society: Sociological Perspectives*, ed. Riley E. Dunlap and Robert J. Brulle. New York: Oxford University Press.
- Egan, Patrick J, David M Konisky and Megan Mullin. 2022. “Ascendant Public Opinion: The Rising Influence of Climate Change on Americans’ Attitudes about the Environment.” *Public Opinion Quarterly* 86(1):134–148.
- Egan, Patrick J. and Megan Mullin. 2012. “Turning Personal Experience into Political Atti-

- tudes: The Effect of Local Weather on Americans' Perceptions about Global Warming." *The Journal of Politics* 74(3):796–809.
- Ekwurzel, B., J. Boneham, M. W. Dalton, R. Heede, R. J. Mera, M. R. Allen and P. C. Frumhoff. 2017. "The rise in global atmospheric CO₂, surface temperature, and sea level from emissions traced to major carbon producers." *Climatic Change* 144(4):579–590.
- Finnegan, Jared J. 2022. "Institutions, Climate Change, and the Foundations of Long-Term Policymaking." *Comparative Political Studies* 55(7):1198–1235.
- Frumhoff, Peter C., Richard Heede and Naomi Oreskes. 2015. "The climate responsibilities of industrial carbon producers." *Climatic Change* 132(2):157–171.
- Gaikwad, Nikhar, Federica Genovese and Dustin Tingley. 2025. "Climate Action from Abroad: Assessing Mass Support for Cross-Border Climate Transfers." *International Organization* 79(1):146–172.
- Gazmararian, Alexander F. 2025. "Sources of Partisan Change: Evidence from the Shale Gas Shock in American Coal Country." *Journal of Politics* .
- Gazmararian, Alexander F. and Dustin Tingley. 2026. "The Political Economy of the Clean Energy Transition."
- Gazmararian, Alexander F. and Helen V. Milner. 2025. "Political Cleavages and Changing Exposure to Global Warming." *Comparative Political Studies* 58(9):1965–1999.
- Gazmararian, Alexander F. and Helen V. Milner. 2026. "Experience and self-interest: Diverging responses to global warming." *American Journal of Political Science* .
- Gilardi, Fabrizio. 2012. Transnational Diffusion: Norms, Ideas, and Policies. In *Handbook of International Relations*, ed. Walter Carlsnaes, Thomas Risse and Beth A. Simmons. United Kingdom: SAGE Publications, Limited pp. 453–477.

- Gourevitch, Peter. 1978. "The Second Image Reversed: The International Sources of Domestic Politics." *International Organization* 32(4):881–912.
- Green, Jessica, Jennifer Hadden, Thomas Hale and Paasha Mahdavi. 2022. "Transition, hedge, or resist? Understanding political and economic behavior toward decarbonization in the oil and gas industry." *Review of International Political Economy* 29(6):2036–2063.
- Hai, Zuhad. 2025. "The Global Politics of Scientific Consensus: Evidence from the Intergovernmental Panel on Climate Change." *International Organization* 79(2):233–256.
- Hale, Thomas. 2020. "Catalytic Cooperation." *Global Environmental Politics* 20(4):73–98.
- Heede, Richard. 2014. "Tracing anthropogenic carbon dioxide and methane emissions to fossil fuel and cement producers, 1854–2010." *Climatic Change* 122(1):229–241.
- Hughes, Llewelyn and Johannes Urpelainen. 2015. "Interests, institutions, and climate policy: Explaining the choice of policy instruments for the energy sector." *Environmental Science & Policy* 54:52–63.
- Kamenica, Emir and Matthew Gentzkow. 2011. "Bayesian Persuasion." *American Economic Review* 101(6):2590–2615.
- Kennard, Amanda. 2020. "The Enemy of My Enemy: When Firms Support Climate Change Regulation." *International Organization* 74(2):187–221.
- Kennard, Amanda. 2025. "The Impacts of Globalization on Political Knowledge: Evidence from Climate Awareness in Africa."
- Kennard, Amanda and Keith E. Schnakenberg. 2023. "Comment: Global Climate Policy and Collective Action." *Global Environmental Politics* 23(1):133–144.

- Keohane, Robert O. and David G. Victor. 2016. "Cooperation and discord in global climate policy." *Nature Climate Change* 6:570–575.
- Kim, Sung Eun, Johannes Urpelainen and Joonseok Yang. 2016. "Electric utilities and American climate policy: lobbying by expected winners and losers." *Journal of Public Policy* 36(2):251–275.
- Kollman, Ken. 1998. *Outside Lobbying: Public Opinion and Interest Group Strategies*. Princeton: Princeton University Press.
- Lee, Tien Ming, Ezra M. Markowitz, Peter D. Howe, Chia-Ying Ko and Anthony A. Leiserowitz. 2015. "Predictors of public climate change awareness and risk perception around the world." *Nature Climate Change* 5(11):1014–1020.
- Li, Hui, Robert P. Berrens, Alok K. Bohara, Hank C. Jenkins-Smith, Carol L. Silva and David L. Weimer. 2004. "Would developing country commitments affect US households' support for a modified Kyoto Protocol?" *Ecological Economics* 48(3):329–343.
- Little, Andrew T. 2023. "Bayesian explanations for persuasion." *Journal of Theoretical Politics* 35(3):147–181.
- Little, Andrew T. 2025. "How to Distinguish Motivated Reasoning from Bayesian Updating." *Political Behavior* .
- Malis, Matt. 2024. "Foreign Policy Appointments." *International Organization* 78(3):501–537.
- Maskin, Eric and Jean Tirole. 2004. "The Politician and the Judge: Accountability in Government." *American Economic Review* 94(4):1034–1054.

- McAllister, Jordan H. and Keith E. Schnakenberg. 2022. “Designing the Optimal International Climate Agreement with Variability in Commitments.” *International Organization* 76(2):469–486.
- Melnick, Justin. 2024. “Domestic Politics in the European Union’s Emissions Trading System: Evidence from Free Allowance Allocation.” *Journal of Political Institutions and Political Economy* 5(4):627–657.
- Melnick, Justin and Alastair Smith. 2025. “Shaming Paris: A Political Economy of Climate Commitments.” *International Organization* 79(2):281–305.
- Mildenberger, Matto. 2020. *Carbon Captured: How Business and Labor Control Climate Politics*. Cambridge: The MIT Press.
- Nachmany, Michal, Sam Fankhauser, Joana Setzer and Alina Averchenkova. 2017. Global trends in climate change legislation and litigation. Technical report Grantham Institute.
- Oreskes, Naomi and Erik M. Conway. 2011. *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Climate Change*. New York: Bloomsbury Publishing.
- Paine, Jack and Scott A. Tyson. 2020. Uses and abuses of formal models in political science. In *The SAGE Handbook of Political Science*, ed. Dirk Berg-Schlosser, Bertrand Badie and Leonardo Morlino. Oxford: Oxford University Press pp. 188–202.
- Ross, Michael L. 2025. “The New Political Economy of Climate Change.” *World Politics* 77(1):155–194.
- Rowan, Sam S. 2025. “From Gridlock to Ratchet: Conditional Cooperation on Climate Change.” *International Organization* 79(2):257–280.

- Schlichting, Inga. 2013. "Strategic Framing of Climate Change by Industry Actors: A Meta-analysis." *Environmental Communication* 7(4):493–511.
- Schnakenberg, Keith E. and Ian R. Turner. 2024. "Formal Theories of Special Interest Influence." *Annual Review of Political Science* 27(Volume 27, 2024):401–421.
- Schultz, Kenneth A. 2005. "The Politics of Risking Peace: Do Hawks or Doves Deliver the Olive Branch?" *International Organization* 59(1):1–38.
- Simmons, Beth A., Frank Dobbin and Geoffrey Garrett. 2006. "Introduction: The International Diffusion of Liberalism." *International Organization* 60(4):781–810.
- Stavins, Robert N. 2011. "The Problem of the Commons: Still Unsettled after 100 Years." *American Economic Review* 101(1):81–108.
- Stokes, Leah C. 2016. "Electoral Backlash against Climate Policy: A Natural Experiment on Retrospective Voting and Local Resistance to Public Policy." *American Journal of Political Science* 60(4):958–974.
- Stokes, Leah C. 2020. *Short Circuiting Policy: Interest Groups and the Battle Over Clean Energy and Climate Policy in the American States*. New York: Oxford University Press.
- Supran, G., S. Rahmstorf and N. Oreskes. 2023. "Assessing ExxonMobil's global warming projections." *Science* 379(6628):eabk0063.
- Supran, Geoffrey. 2022. "Three shades of green(washing): Content analysis of social media discourse by European oil, car, and airline companies."
- Supran, Geoffrey and Naomi Oreskes. 2021. "Rhetoric and frame analysis of ExxonMobil's climate change communications." *One Earth* 4(5):696–719.

- Tarar, Ahmer. 2006. “Diversionary Incentives and the Bargaining Approach to War.” *International Studies Quarterly* 50(1):169–188.
- Tingley, Dustin and Michael Tomz. 2014. “Conditional Cooperation and Climate Change.” *Comparative Political Studies* 47(3):344–368.
- Urpelainen, Johannes and Alice Tianbo Zhang. 2022. “Electoral Backlash or Positive Reinforcement? Wind Power and Congressional Elections in the United States.” *The Journal of Politics* 84(3):1306–1321.
- Urpelainen, Johannes and Thijs Van de Graaf. 2018. “United States non-cooperation and the Paris agreement.” *Climate Policy* 18(7):839–851.
- Voeten, Erik. 2025. “The Energy Transition and Support for the Radical Right: Evidence from the Netherlands.” *Comparative Political Studies* 58(2):394–428.
- Williams, Emily L., Sydney A. Bartone, Emma K. Swanson and Leah C. Stokes. 2022. “The American electric utility industry’s role in promoting climate denial, doubt, and delay.” *Environmental Research Letters* 17(9):094026.